Modelling Sovereign Credit Spreads with International Macro Factors: The Case of Brazil 1998-2009.

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Abstract

This paper develops a macro-finance model of the Brazilian economy and its sovereign debt markets that allows for domestic and international macroeconomic influences as well as swings in investor confidence. It finds significant evidence of common trends in the US and Brazilian economies and bond markets as well as spillover effects from US inflation and business cycles to the Brazilian economy. The US Fed Funds rate influences Brazilian sovereign spreads, as do Brazilian inflation and policy rates. The Brazilian common risk factor dominates the behavior of the spreads during periods of crisis and we find that it also affects the level and volatility of macroeconomic variables. These results suggest that the macro-finance approach could throw light upon the behavior of other emerging economies and markets.

Key words: affine term structure model, macro finance, sovereign credit spread, international spillover, macroeconomic volatility.

JEL Codes: G12,F41,F34,E43

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1 Introduction

This paper develops a macro-finance model of the US and Brazilian economies and the markets for Brazilian sovereign debt (US dollar-denominated Brazilian government bonds) over the period November 1998 to December 2009. This was a period of political stability under the Cardoso and Lula governments, during which Brazil (along with the other 'BRICs', Russia, India and China) emerged as a leading world economy. It followed the emerging market crisis of 1998 but included the crises of 1999 and 2002 as well as the global financial crisis of 2008.

There is a large outstanding literature on the effects of macroeconomic and financial factors on sovereign spreads (the differences between yields on sovereign bonds and US Treasuries). Most of these studies analyze the effect of a domestic macroeconomic and external liquidity conditions on these spreads (Eichengreen and Mody (1998), Min (1998) . Ferrucci (2003) and Baek, Bandopadhyaya, and Du (2005)). Others find that contagion spillovers from other emerging markets (Mauro, Sussman, and Yafeh (2002), Ciarlone, Piselli, and Trebeschi (2009), IMF (1999) and Claessens and Forbes (2001)) as well as global financial factors (Masson and Mussa (1995), and Arora and Cerisola (2001)) are also very significant, often dominating the effect of macroeconomic variables. However, these studies do not use the arbitrage-free pricing approach found elsewhere in the finance literature. Nor do they consider the possible effects of swings in investor confidence on the domestic economy.

In this paper we use an arbitrage-free macro-finance framework that allows bond yields to reflect macroeconomic variables as well as latent variables representing spillovers from other emerging markets, global influences and other financial market factors. It also allows these factors to influence the domestic economy. This approach was pioneered by Ang and Piazzesi (2003) and has been developed by Rudebusch and Wu (2003), Dewachter and Lyrio (2006)

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and many others. Most macro-finance studies have adopted a closed economy model, but we follow Spencer and Liu (2010) by developing an open economy model that allows for external effects as well as macroeconomic spillovers from the US to Brazil. Normally, these models also assume a homoscedastic error structure, but in this study we allow for 'square root volatility' as formulated by (Cox, Ingersoll, and Ross (1985), henceforth CIR, and Sun (1992)).

We use a reduced-form framework[1] to model Brazilian sovereign bond spreads. This approach treats default as an unexpected event that occurs with a hazard rate $h(t)$, which can be modelled like the spot rate $r(t)$ in a standard affine term structure model. It has successfully replicated the size of credit spreads in the corporate markets (Jarrow, Lando, and Turnbull (1997), Lando (1998) and Duffie and Singleton (1999)). Liquidity, counterparty and other market-specific risks can be handled in a similar way. Duffie, Pedersen, and Singleton (2003) and Pan and Singleton (2008) use this framework to model sovereign spreads. However, these reduced-form models all use latent variables that can only be interpreted as confidence indicators or perhaps by *ex post* alignment with macroeconomic variables. Our model also incorporates macroeconomic variables, allowing us to distinguish their effects from those of unobservable 'confidence' and other effects.

Our estimation period includes a period of high sovereign spreads and macroeconomic volatility, seen until 2004, allowing us to assess the effect of confidence factors on both spreads and the domestic economy. However, it extends to December 2009, including subsequent observations for which investor confidence in Brazil was high and macroeconomic volatility

[1] The alternative would be to use a structural model in which default is triggered as a solvency variable (like the value of a firm in the case of a corporate borrower) hits a threshold value (Merton (1974), Leland (1994), Leland and Toft (1996), Longstaff and Schwartz (1995)). However, the solvency variable is not well defined for a sovereign borrower and the threshold would be hard to specify. It is also difficult to explain the size of short spreads using the structural approach.

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low, allowing us to identify significant macroeconomic effects on yield spreads and volatility. This data set also encompasses the credit crunch that began in August 2007 and the ensuing US recession. This caused much larger fluctuations in US macroeconomic variables than seen in earlier years, making it easier to calibrate the likely spillover effects. Granger Causality tests suggest that these fluctuations impacted the Brazilian economy and markets significantly, motivating the open economy specification of this paper.

Our econometric model consists of two sub-systems. The first is a standard macro-finance model of the US economy and Treasury bond market. This is based on the Central Bank Model (CBM) developed by Svensson (1999), Rudebusch (2002), Smets (1999) and others, which represents the behavior of the macroeconomy in terms of the US output gap (g_t^*), inflation (π_t^*) and the short term interest rate (r_t^*). It also includes two latent variables reflecting real (z_t^*) and nominal (f_t^*) factors, which are interpreted as 'common' or 'global' factors because they also affect Brazil. The second sub-system is a CBM-based model of the Brazilian economy and bond markets represented by the output gap (g_t), inflation (π_t) and short term interest rates (r_t) as well as sovereign bond spreads.

This model allows for common global trends as well as spillovers from US macroeconomic variables to Brazil. It includes two additional latent variables. The first factor (f_t), denoted as the common risk factor, is designed to reflect overseas confidence in the Brazilian economy (including any spillover effects from other emerging markets). We find that it also has significant effects on both the level and volatility of the Brazilian macroeconomic variables. The sovereign bond market spreads reflect the common risk indicator (f_t) plus a bond-market-specific indicator (q_t) that allows for residual default and liquidity risks. These various latent variables are modelled using the Kalman filter and the two sub-models are estimated simultaneously using

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Maximum Likelihood.

We adopt a general-to-specific approach to identifying the common factor and spillover effects from the US. We start with an unrestricted spillover matrix in model M3 and then in model M2 restrict this in line with our classification of variables and interactions as real or nominal. Elimination of the remaining insignificant effects then gave the final specification M1 which is the focus of our discussion. The closed economy model M0 is strongly rejected against these alternatives. Model M1 confirms the importance of a common trend (z_t^*) in real interest rates. It also provides evidence of a common nominal trend (f_t^*). However, this is harder to interpret, since it reflects both inflationary effects associated with oil prices and 'flight to quality' effects seen in world bond markets. In addition to these global effects, the US output gap impacts the Brazil business cycle and there is also evidence of an inflationary spillover.

We find that although the US Fed funds rate has no significant effect on the Brazilian economy it does affect the sovereign spreads, as do the Brazilian inflation and policy interest rates. The confidence indicator (f_t) dominates the pre-2005 period. However, it falls back sharply after the crisis of 2002 and is then very stable, allowing macroeconomic influences to be found. We find that the f_t has a significant effect on macroeconomy volatility, consistent with the stability exhibited over the latter part of the period. Surprisingly in view of the evidence for 'square root volatility' in US term structure models we find no significant evidence of this in either US macroeconomic or yield data. This also seems to reflect the characteristics of this data set. While long runs of US data suggest that the secular up-trend in rates of inflation and interest until 1979-80 and their subsequent decline are associated with movements in volatility there is no obvious trend in mean rates or volatility here. Indeed the sharpest variations in US macroeconomic variables occur towards the end of the period when inflation and yields are

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relatively low.

The remainder of the paper is organized as follows. Section 2 lists some of the main developments over the estimation period 1998-2009 and then discusses the data and the preliminary tests. Section 3 describes the model specification and Section 4 reviews the empirical results. Section 5 provides our conclusion.

2 Key developments, data and model design

This section outlines the key political and economic developments seen during our estimation period. It then reports the model data sources and the results of a preliminary specification search, used to inform the model framework in the next section.

2.1 Data sources

The model is estimated using monthly macro and yield data for the period November 1998 to December 2009, which was an eventful period in Brazil's political and economic history. It began under President Fernande Henrique Cardoso with a move from fixed to floating exchange rates in January 1999 and to inflation targeting the following July. It includes the periods of uncertainty seen in 1999 and again in 2002 following the Argentinian default and the election of President Luis Inacio Lula da Silva. When Sr. da Silva came to power, Brazil had already received the largest loan ever given by the IMF and the markets expected him to default. However, he tightened fiscal policy instead, making it much tighter than the IMF had demanded. Confidence began to return towards the end of the year, ushering in a period of economic growth and stability. This placed Brazil in a good position to withstand the effect of the credit

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crunch that began in the US in August 2007, when a liquidity crisis in the Brazilian markets with potentially serious economic consequences was effectively dealt with by a reduction in bank reserve requirements. Improvements in macroeconomic policy and behavior allowed the country to ride the reduction in world trade relatively well and then to recover remarkably quickly.

Turning to the data sources, we use for π_t the core IPCA inflation data[2] calculated by the Banco Central do Brazil (BCB) using the symmetric trimmed-mean method. The US inflation rate π_t^* is the annual percentage change in the all items CPI (Source: US Bureau of Labor Statistics). The Brazilian output gap, g_t, is obtained by de-trending IBGE's industrial production index (seasonally and inflation adjusted) using the Hodrick-Prescott filter (with $\lambda = 14400$). The US output gap, g_t^*, is obtained by similarly de-trending a monthly GDP estimator provided by E-forecasting.com. The Brazil short rate, r_t, is the Brazil Treasury Bill rate taken from IMF International Financial Statistics, while the US short rate r_t^* is the 1-month US Treasury Bill rate published by the Federal Reserve. We regard these as policy rates. These macroeconomic data are shown in Figure 1 and summary statistics are reported in Table 1. The Brazilian sovereign yields are the discount bond equivalent rates (i.e. spot rate) constructed using the Nelson and Siegel model with US dollar dominated Brazilian coupon-bond data obtained from Bloomberg[3]. The US discount bond equivalent rates are provided by the New York Federal Reserve Bank. These data are shown in Figures 2 and table 1 reports the summary statistics.

[2] Core IPCA inflation is a measure of IPCA inflation that purges some of the elements from the IPCA in order to achieve a more accurate view of the underlying trend of price changes in the economy. The IPCA (Broad Consumer Price Index) is the most important price index from the standpoint of macroeconomic policy because it is the consumer price index that is used in the country's inflation targeting regime adopted in June 1999.

[3] These are semi-annual coupon bonds dominated in US dollars that are issued by the Brazilian government and traded in the international market.

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2.2 Preliminary tests and model design

The ADF tests reported in Table 1 suggest that the US and Brazilian output gaps are stationary, while inflation and interest rates are non-stationary. Surprisingly, US and Brazilian real interest rates also appear to be non-stationary in this data set. Further tests show that there are 3 common trends underlying the 4 non-stationary variables. This suggested the use of a non-stationary 'real interest rate factor' affecting both the Brazil and US macro models. We also employ a nominal 'financial factor' to capture any global inflation trends and an orthogonal 'financial factor' for the Brazil macro and spread models. Further Granger causality tests suggested that the US output gap and inflation rates affected their Brazilian counterparts. Informed by these results, our modelling strategy was to specify a general model in which all US latent and observable variables could potentially affect the Brazilian economy and sovereign yields and then sequentially eliminate insignificant spillovers. Finally, the dynamic structure was informed by Bayesian Information Criterion (BIC) tests on the two basic macro VAR models (excluding latent variables and spillovers). These tests are reported in table 2. They suggest a second order model is needed to model the US macro data, while a first order model is adequate for Brazil.

3 The model framework

This section describes the model structure, which reflects the preliminary results reported in the previous section. The two macro systems are specified as Kalman Vector Autoregression KVAR(N^*) models under the real world or state density probability measure $\mathcal{P}$, where N^* is the order of the lag length. The companion or state space form is given in Appendix A.1.

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The financial market models adjust these KVAR structures using the US dollar prices of risk defined in Appendix A.2 to represent the dynamics under the risk neutral probability measure $\mathcal{Q}$. In this arbitrage-free affine pricing framework, the discount bond yields are then linear in the state variables.

3.1 The US macro KVAR

The BIC test shown in table 2 indicates that a second order difference system was appropriate to describe the US macroeconomic dynamics ($N^* = 2$). This gives:

$$\mathbf{x}_t^* = \boldsymbol{\kappa}^* + \boldsymbol{\phi}_z^* z_t^* + \boldsymbol{\phi}_f^* f_t^* + \boldsymbol{\Phi}_1^* \mathbf{x}_{t-1}^* + \boldsymbol{\Phi}_2^* \mathbf{x}_{t-2}^* + \mathbf{w}_t^* \tag{1}$$

$$\mathbf{w}_t^* = \mathbf{G}^* \mathbf{D}_{t-1}^{x*} \boldsymbol{\epsilon}_t^{x*},\ \mathbf{D}_{t-1}^{x*} = \mathbf{D}_0^{x*} + \mathbf{D}_1^{x*} \sqrt{f_{t-1}^*},\ \boldsymbol{\epsilon}_t^{x*} \sim N(\mathbf{0}, \mathbf{I})$$

where $\mathbf{x}_t^* = \{g_t^*, \pi_t^*, r_t^*\}'$ is the observed US macro vector including output, inflation and short rate, and $\mathbf{w}_t^*$ is a 3×1 error vector; $\mathbf{D}_0^{x*}$ and $\mathbf{D}_1^{x*}$ are both 3 by 3 diagonal matrices with positive diagonal elements and $\mathbf{G}^*$ is a 3×3 lower triangular matrix with unit diagonal. The real factor is a unit-root Vasicek-type process under $\mathcal{P}$:

$$z_t^* = z_{t-1}^* + \delta^{z*} \epsilon_t^{z*}, \epsilon_t^{z*} \sim N(0,1)$$

The nominal factor f_t^* is non-negative and follows a CIR-type process under $\mathcal{P}$:

$$f_t^* = \xi^{f*} f_{t-1}^* + \delta^{f*} \sqrt{f_{t-1}^*} \epsilon_t^{f*}; \tag{2}$$

$$\epsilon_t^{f*} \sim N(0,1); \left|\xi^{f*}\right| \leq 1.$$

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The central tendency $\tilde{\mathbf{x}}_t^*$ of the US macro vector is driven by the two factors:

$$\tilde{\mathbf{x}}_t^* = E_t\left(\lim_{m\to\infty} \mathbf{x}_{t+m}^* | z_t^*, f_t^*\right) = \boldsymbol{\varphi}^* + \mathbf{R}_1^* z_t^* + \mathbf{R}_2^* f_t^* \tag{3}$$

$\tilde{\mathbf{x}}_t^*$ is the equilibrium conditional expectation under the state price density measure $\mathcal{P}$ given the current values of the factors. We incorporate the cointegration constraints identified in our preliminary regression analysis using the restrictions: $\mathbf{R}_1^* = (0\ 0\ 1)'$, $\mathbf{R}_2^* = (0\ 1\ 1)'$ and $\boldsymbol{\varphi}^* = (0,\ \varphi_1^*,\ \varphi_2^*)'$. These restrictions make the output gap mean-revert to zero and allow us to interpret $\varphi_1^* + f_t^*$ and $\varphi_2^* + z_t^* + f_t^*$ as the inflation and nominal interest rate central tendencies. The US macro model (1) also implies the relationship:

$$\tilde{\mathbf{x}}_t^* = \left(\mathbf{I} - \boldsymbol{\Phi}_1^* - \boldsymbol{\Phi}_2^*\right)^{-1}\left(\boldsymbol{\kappa}^* + \boldsymbol{\phi}_z^* z_t^* + \boldsymbol{\phi}_f^* f_t^*\right) \tag{4}$$

Comparing (3) and (4) constraints the parameters of (1): $\boldsymbol{\kappa}^* = \left(\mathbf{I} - \boldsymbol{\Phi}_1^* - \boldsymbol{\Phi}_2^*\right)\boldsymbol{\varphi}^*$; $\boldsymbol{\phi}_z^* = \left(\mathbf{I} - \boldsymbol{\Phi}_1^* - \boldsymbol{\Phi}_2^*\right)\mathbf{R}_1^*$; $\boldsymbol{\phi}_f^* = \left(\mathbf{I} - \boldsymbol{\Phi}_1^* - \boldsymbol{\Phi}_2^*\right)\mathbf{R}_2^*$.

3.2 The Brazilian macro KVAR

Our model allows the observed Brazilian macro vector $\mathbf{x}_t = \{g_t, \pi_t, r_t\}'$ to be affected by its lagged values, the common financial risk factor f_t and the US macro variables. The BIC test reported in Table 2 indicates that a first order difference system is appropriate in this case ($N = 1$):

$$\mathbf{x}_t = \boldsymbol{\kappa} + \boldsymbol{\Theta}\mathbf{X}_t^* + \boldsymbol{\phi} f_t + \boldsymbol{\Phi}_1 \mathbf{x}_{t-1} + \mathbf{w}_t \tag{5}$$

$$\mathbf{w}_t = \mathbf{G}\mathbf{D}_{t-1}^x \boldsymbol{\epsilon}_t^x, \mathbf{D}_{t-1}^x = \mathbf{D}_0^x + \mathbf{D}_1^x \sqrt{f_{t-1}}, \boldsymbol{\epsilon}_t^x \sim N(\mathbf{0}, \mathbf{I}),$$

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where $\mathbf{X}_t^* = \{z_t^*, f_t^*, \mathbf{x}_t^{*\prime}, \mathbf{x}_{t-1}^{*\prime}\}'$; $\mathbf{\Theta X}_t^* = \boldsymbol{\theta}_z z_t^* + \boldsymbol{\theta}_f f_t^* + \boldsymbol{\theta}_{x1}\mathbf{x}_t^* + \boldsymbol{\theta}_{x2}\mathbf{x}_{t-1}^*$; the error term $\mathbf{w}_t$ is a 3×1 vector; $\mathbf{D}_0^x$, $\mathbf{D}_1^x$ are both diagonal matrices with positive diagonal elements; $\mathbf{G}$ is a lower triangular matrix with unit diagonal. Under $\mathcal{P}$ the financial risk factor f_t follows another CIR process of form shown below:

$$f_t = \xi^f f_{t-1} + \delta^f \sqrt{f_{t-1}}\epsilon_t^f; \tag{6}$$
$$\epsilon_t^f \sim N(0,1); \left|\xi^f\right| \leq 1.$$

The central tendency of the Brazil macro vector, $\tilde{\mathbf{x}}_t$, can be expressed as a function of the US variables and the Brazil common financial risk factor as:

$$\tilde{\mathbf{x}}_t = E_t\left(\lim_{m\to\infty} \mathbf{x}_{t+m} | \mathbf{X}_t^*, f_t\right) = \boldsymbol{\varphi} + \mathbf{\Xi X}_t^* + \mathbf{R} f_t \tag{7}$$

Solving (5) also implies the relationship:

$$\tilde{\mathbf{x}}_t = \left(\mathbf{I} - \mathbf{\Phi}_1\right)^{-1}\left(\boldsymbol{\kappa} + \mathbf{\Theta X}_t^* + \boldsymbol{\phi} f_t\right) \tag{8}$$

and comparing (7) and (8) gives the constraints: $\boldsymbol{\kappa} = \left(\mathbf{I} - \mathbf{\Phi}_1\right)\boldsymbol{\varphi}$; $\mathbf{\Theta} = \left(\mathbf{I} - \mathbf{\Phi}_1\right)\mathbf{\Xi}$; $\boldsymbol{\phi} = \left(\mathbf{I} - \mathbf{\Phi}_1\right)\mathbf{R}$.

In the baseline closed economy model M0, the world economy does not affect Brazil ($\mathbf{\Xi} = \mathbf{0}_{3,8}$) and the financial common risk factor f_t determines the central tendency of $\mathbf{x}_t$. We employ the restrictions: $\mathbf{R} = (0,\, \upsilon,\, \upsilon)'$ and $\boldsymbol{\varphi} = (0,\, \varphi_1,\, \varphi_2)'$. These make the output gap mean-revert to zero and allow us to interpret $\upsilon f_t + \varphi_1$ and $\upsilon f_t + \varphi_2$ as the inflation and nominal interest rate central tendencies. In a more general model setup (where $\boldsymbol{\theta} \neq \mathbf{0}$), the US variables $\mathbf{X}_t^*$

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can also affect Brazilian inflation and interest rate central tendencies as we now explain.

In model M3 the spillover effects are unrestricted. In model M2 we apply restrictions (similar to those on $\mathbf{R}$) on $\mathbf{\Xi}$ to identify the real and nominal spillover effects. We first restrict $\mathbf{\Xi}$ so that the Brazilian output gap only depends upon the US gap in the long run: by setting all elements of the first row of $\mathbf{\Xi}$ apart from χ_{13} and χ_{16} to zero (where χ_{ij} is the element in the i-th row and j-th column of $\mathbf{\Xi}$). Next we use the restriction $\chi_{21} = -(\chi_{25} + \chi_{28})$ to ensure that the long run effect of a change in the US real interest rate (a unit increase in z_t^*, r_t^* and r_{t-1}^*) affects the Brazilian interest rate but not the inflation rate. So the long run effects matrix $\mathbf{\Xi}$ in M2 is defined as:

$$\mathbf{\Xi} = \begin{pmatrix} 0 & 0 & \chi_{13} & 0 & 0 & \chi_{16} & 0 & 0 \\ -(\chi_{25} + \chi_{28}) & \chi_{22} & \chi_{23} & \chi_{24} & \chi_{25} & \chi_{26} & \chi_{27} & \chi_{28} \\ \chi_{31} & \chi_{32} & \chi_{33} & \chi_{34} & \chi_{35} & \chi_{36} & \chi_{37} & \chi_{38} \end{pmatrix}. \tag{9}$$

Further work on model M2 showed that several of the these spillover parameters were poorly determined. Sequential elimination of parameters with a $t-$value less than unity gave the model reported as M1.

Finally, we use another orthogonal latent factor to pick up the residual risk factors in the Brazilian sovereign bond markets. This is similar to the dynamics of f_t/f_t^* under P,

$$q_t = \kappa^q + \xi^q q_{t-1} + \delta^q \sqrt{q_{t-1}} \epsilon_t^q; \tag{10}$$

$$\epsilon_t^q \sim N(0,1); |\xi^q| \leq 1.$$

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We discuss the factor in more details in section 4.

3.3 The companion form

The companion or state space form describes the US and Brazilian macro models as a single first order difference equation system by stacking them as:

$$\mathbf{X}_t = \mathbf{K} + \mathbf{\Phi}\mathbf{X}_{t-1} + \mathbf{W}_t \tag{11}$$

$where$:

$$\mathbf{W}_t = \mathbf{C}\mathbf{D}_{t-1}\boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_t \sim N(\mathbf{0}, \mathbf{I}); \mathbf{\Omega}_t = \mathbf{C}\mathbf{D}_{t-1}^2\mathbf{C}'; \tag{12}$$

$$\mathbf{D}_t^2 = \mathbf{D}_0^2 + \mathbf{D}_1^2\mathbf{J}_{f*}\mathbf{X}_t + \mathbf{D}_2^2\mathbf{J}_f\mathbf{X}_t + \mathbf{D}_3^2\mathbf{J}_{fb}\mathbf{X}_t + \mathbf{D}_4^2\mathbf{J}_{fc}\mathbf{X}_t \tag{13}$$

where $\mathbf{X}_t = \left(z_t^*, f_t^*, \mathbf{x}_t^{*\prime}, \mathbf{x}_{t-1}^{*\prime}, f_t, \mathbf{x}_t', q_t\right)'$; $\mathbf{J}_{f*}, \mathbf{J}_f$, and $\mathbf{J}_q$ are row selection vectors which extract f_t^*, f_t, and q_t respectively from $\mathbf{X}_t; \mathbf{D}_t$ is a diagonal matrix. The parameter matrices $\mathbf{K}$, $\mathbf{\Phi}$, $\mathbf{C}$, $\mathbf{D}_0, \mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3$ are defined in Appendix A.1.

The macro models are specified and estimated under the measure $\mathcal{P}$. Financial models use a risk-neutral probability measure such as $\mathcal{Q}$, which has the effect of adjusting these probabilities so that all assets have the same expected US *dollar* rate of return. This adjustment is made by changing the parameters of (11) to get a congruent first-order process:

$$\mathbf{X}_t = \mathbf{K}^Q + \mathbf{\Phi}^Q\mathbf{X}_{t-1} + \mathbf{W}_t^Q; \mathbf{W}_t^Q \sim N(\mathbf{0}, \mathbf{\Omega}_t). \tag{14}$$

where:

$$\mathbf{K}^Q = \mathbf{K} - \mathbf{C}\mathbf{D}_0^2\mathbf{\Lambda}_1 \tag{15}$$

$$\mathbf{\Phi}^Q = \mathbf{\Phi} - \mathbf{C}\mathbf{D}_1^2\mathbf{\Lambda}_1\mathbf{J}_{f*} - \mathbf{C}\mathbf{D}_2^2\mathbf{\Lambda}_1\mathbf{J}_f - \mathbf{C}\mathbf{D}_3^2\mathbf{\Lambda}_1\mathbf{J}_{fb} - \mathbf{C}\mathbf{\Lambda}_2 \tag{16}$$

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The parameters $\mathbf{\Lambda}_1$ and $\mathbf{\Lambda}_2$ incorporate the 'prices of risk' associated with the US and Brazilian variables and are specified in Appendix A.2.

3.4 Bond pricing models

In the affine model framework, the US dollar default free discount (or zero coupon) bond prices are log-linear in the state variables. The price $P_{\tau,t}$ at time t of a default-free dollar payment at time $t+\tau$ is:

$$-\ln P_{\tau,t} = \eta^*_\tau + \mathbf{\Psi}^*_\tau \mathbf{X}_t \tag{17}$$

where:

$$\begin{aligned} \eta^*_\tau &= \eta^*_{\tau-1} + \mathbf{\Psi}^*_{\tau-1}\mathbf{K}^Q - \frac{1}{2}\mathbf{\Psi}^*_{\tau-1}\mathbf{C}\mathbf{D}_0^2\mathbf{C}'\mathbf{\Psi}^{*\prime}_{\tau-1} \\ \mathbf{\Psi}^*_\tau &= \mathbf{\Psi}^*_1 + \mathbf{\Psi}^*_{\tau-1}\mathbf{\Phi}^Q - \frac{1}{2}\mathbf{\Psi}^*_{\tau-1}\mathbf{C}\mathbf{D}_1^2\mathbf{C}'\mathbf{\Psi}^{*\prime}_{\tau-1}\mathbf{J}_{f*} \end{aligned} \tag{18}$$

with initial conditions: $\eta^*_1 = 0, \mathbf{\Psi}^*_1 = \mathbf{J}^*_r$, where $\mathbf{J}^*_r$ is a selection vector that extracts r^*_t from $\mathbf{X}_t : r^*_t = \mathbf{J}^*_r\mathbf{X}_t$.

To price a *defaultable* discount US dollar denominated Brazilian government bond $V^b_{\tau,t}$, we follow the reduced-form approach in Duffie, Pedersen, and Singleton (2003), and rewrite their formula in the discrete-time format:

$$V^b_{\boldsymbol{\tau},t} = E^Q_t\left[\exp\left(-\sum_{i=t}^{t+\tau-1} R_i\right)\right] \tag{19}$$

where $R_t = r^*_t + s_t$ is the 'default and liquidity adjusted' short rate and s_t is the one-period short credit spread.

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The short spread s_t is assumed to be affected by the Brazilian common risk factor f_t, Brazilian macro variables $\mathbf{x}_t$, the US interest rate r_t^* and a sovereign bond market specific factor q_t.

$$s_t = f_t + \gamma^r r_t^* + \gamma^x \mathbf{x}_t + q_t = \mathbf{J}_s \mathbf{X}_t \tag{20}$$

$$where : \mathbf{J}_s = \{\mathbf{0}_{1,4}, \gamma^r, \mathbf{0}_{1,3}, 1, \gamma^x, 1, 0\}$$

The price of the US-dollar denominated discount Brazilian sovereign bond is then given by

$$-\ln V_{\tau,t}^b = \eta_\tau + \mathbf{\Psi}_\tau \mathbf{X}_t \tag{21}$$

where:

$$\eta_\tau = \eta_{\tau-1} + \mathbf{\Psi}_{\tau-1}\mathbf{K}^Q - \frac{1}{2}\mathbf{\Psi}_{\tau-1}\mathbf{C}\mathbf{D}_0^2\mathbf{C}'\mathbf{\Psi}'_{\tau-1} \tag{22}$$

$$\mathbf{\Psi}_\tau = \mathbf{\Psi}_1 + \mathbf{\Psi}_{\tau-1}\mathbf{\Phi}^Q - \frac{1}{2}\mathbf{\Psi}_{\tau-1}\mathbf{C}\mathbf{D}_1^2\mathbf{C}'\mathbf{\Psi}'_{\tau-1}\mathbf{J}_{f*} - \frac{1}{2}\mathbf{\Psi}_{\tau-1}\mathbf{C}\mathbf{D}_2^2\mathbf{C}'\mathbf{\Psi}'_{\tau-1}\mathbf{J}_f - \frac{1}{2}\mathbf{\Psi}_{\tau-1}\mathbf{C}\mathbf{D}_3^2\mathbf{C}'\mathbf{\Psi}'_{\tau-1}\mathbf{J}_q$$

with: $\eta_1 = 0, \mathbf{\Psi}_1 = \mathbf{J}_s + \mathbf{J}_r^*$.

Dividing by maturity and adding an error term gives the two empirical yield models: the US yield model is $y_{\tau,t}^* = -\ln P_{\tau,t}/\tau = \alpha_\tau^* + \boldsymbol{\beta}_\tau^* \mathbf{X}_t + e_{\tau,t}^*$, and the Brazilian yield model is $y_{\tau,t} = -\ln V_{\tau,t}^b/\tau = \alpha_\tau + \boldsymbol{\beta}_\tau \mathbf{X}_t + e_{\tau,t}$, where $y_{\tau,t}^*$ and $y_{\tau,t}$ are the yields of the τ-period US and Brazilian (US dollar) discount bonds respectively at time t; $e_{\tau,t}^*$ and $e_{\tau,t}$ are measurement errors; $\alpha_\tau^* = \eta_\tau^*/\tau$, $\boldsymbol{\beta}_\tau^* = \mathbf{\Psi}_\tau^*/\tau$, $\alpha_\tau = \eta_\tau/\tau$, $\boldsymbol{\beta}_\tau = \mathbf{\Psi}_\tau/\tau$; $e_{\tau,t}^* \sim N(0, \sigma_\tau^{*2})$, $e_{\tau,t} \sim N(0, \sigma_\tau^2)$; $\sigma_\tau^* > 0, \sigma_\tau > 0$. The slope coefficients $\boldsymbol{\beta}_\tau^*, \boldsymbol{\beta}_\tau$ are known as the 'factor loadings' in this literature.

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Stacking the two yield models for M different maturities gives:

$$\mathbf{y}_t^* = \boldsymbol{\alpha}^* + \boldsymbol{\beta}^* \mathbf{X}_t + \mathbf{e}_t^*, \mathbf{e}_t^* \sim N(\mathbf{0}, \boldsymbol{\Sigma}^*) \tag{23}$$

$$\mathbf{y}_t = \boldsymbol{\alpha} + \boldsymbol{\beta} \mathbf{X}_t + \mathbf{e}_t, \mathbf{e}_t \sim N(\mathbf{0}, \boldsymbol{\Sigma}) \tag{24}$$

where $\mathbf{y}_t^* = \{y_{m*_1,t}^*, y_{m*_2,t}^* ... y_{m*_{M*},t}^*\}'$; $\mathbf{y}_t = \{y_{m_1,t}, y_{m_2,t} ... y_{m_M,t}\}'$. The Brazil sovereign bond spreads are thus given by

$$s_{\tau,t} = y_{\tau,t} - y_{\tau,t}^* = \alpha_\tau - \alpha_\tau^* + (\boldsymbol{\beta}_\tau - \boldsymbol{\beta}_\tau^*) \mathbf{X}_t, \quad \tau = m_1, ..., m_M \tag{25}$$

For the US bonds, we have 1, 2, 3, 5, 7, and 10 year maturity spot rates. So we have $\{m_1^*, m_2^*, m_3^*, ..., m_{M*}^*\} = \{12, 24, 36, 60, ..., 120\}$ measured in calendar months and define: $\boldsymbol{\beta}^* = \{\boldsymbol{\beta}_{12}^{*\prime}, \boldsymbol{\beta}_{24}^{*\prime}, ...\boldsymbol{\beta}_{120}^{*\prime}\}$, $\mathbf{e}_t^* = diag\{e_{12,t}^*, e_{24,t}^*, ..., e_{120,t}^*\}$, and $\boldsymbol{\Sigma}^* = diag\{\sigma_{12}^{*2}, \sigma_{24}^{*2}, ..., \sigma_{120}^{*2}\}$. For the Brazilian sovereign bonds, we have 2, 3, 5, 7, and 10 year maturity spot rates, have $\{m_1, m_2, m_3, ..., m_M\} = \{24, 36, 60, ..., 120\}$ measured in calendar months and define: $\boldsymbol{\beta} = \{\boldsymbol{\beta}_{24}', \boldsymbol{\beta}_{36}', ...\boldsymbol{\beta}_{120}'\}$, $\mathbf{e}_t = diag\{e_{24,t}, e_{36,t}, ..., e_{120,t}\}$, $\boldsymbol{\Sigma} = diag\{\sigma_{24}^2, \sigma_{36}^2, ..., \sigma_{120}^2\}$.

4 The empirical model

We estimate the US & Brazilian macro KVARs (Equations (1) and (5)); the US and Brazilian yield specifications (23) and (24) simultaneously using the Extended Kalman Filter[4] and Maximum Loglikelihood Estimation method (See Appendix A.3). The loglikelihood is optimized numerically using the Nelder-Mead Simplex algorithm (implemented in the MatLab *fminsearch*

[4]The EKF gives the optimal linear estimator. Stanton (1997) suggests that it is much more robust than other methods like Efficient Method of Moments, which are moreover highly computer-time intensive.

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function). The baseline closed economy model M0 has 88 parameters[5]. The M1, M2 and M3 models which allow for macroeconomic spillovers have 96, 105 and 112 parameters respectively. Table 3 reports the optimized loglikelihood of each model. Since these models are nested, the validity of the closed economy and other sets of restrictions can be tested using Likelihood Ratio (LR) tests. These show that the spillovers are significant on this data set: M0 is rejected against all of these alternatives. This table also shows that the restrictions embodied in M1 and M2 are accepted at the conventional 95% level.

4.1 Empirical results

In view of these findings we now focus on the estimates obtained from our final model: M1[6]. Tables 4a, 4b, and 4c report the parameter estimates and $t-$values. The residuals from the macro and yield variables are shown in Figure 1 and 2. The estimated common and country specific factors are shown in Figure 3. The real interest rate factor z^* clearly reflects movements in US monetary policy, increasing during the dot-com boom at the turn of the decade and then falling sharply under the Greenspan and Bernanke stimuli of and 2001 and 2007. The nominal factor f^* is more difficult to interpret. It partly reflects fluctuations in oil prices, which fell back to an all time low in real terms in 1998 and peaked at an all time high in the Autumn of 2008 before falling back sharply in late 2008. It acts like an adverse supply shock, depressing output initially and increasing inflation and short term interest rates. However these shocks are associated with developments like the 'flight to quality' that drove long term US Treasury yields to their low point in late 2008. The model identifies these two factors as common

[5] With ξ^{f*}, ξ^f, and ξ^q bounded at unity, these are: $\varphi^*(2)$, $\mathbf{\Phi}_1^*(9)$, $\mathbf{\Phi}_2^*(9)$, $\mathbf{G}^*(3)$, δ^{z*}, δ^{f*}, $\mathbf{D}_0^{x*}(3)$, $\mathbf{D}_1^{x*}(3)$, $\varphi(2)$, $\mathbf{\Phi}_1(9)$, $\mathbf{G}(3)$, υ, δ^f, $\mathbf{D}_0^x(3)$, $\mathbf{D}_1^x(3)$, κ^q, δ^q, $\mathbf{\Lambda}_1(10)$, $\mathbf{\Lambda}_2(19)$, γ^r, $\boldsymbol{\gamma}^x(3)$.

[6] Estimates for the other models are available upon request from the authors.

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non-stationary trends, which influence both the US and Brazilian economies and markets.

The remaining factors are used to model the market confidence, liquidity and other market-specific factors affecting the Brazilian economy, and sovereign debt markets. The common risk factor f is a basic Brazilian 'confidence' factor and in this model it also has a significant effect on both the mean and the volatility of Brazilian inflation and interest rates (the latter effects are indicated by the t-values of $D^{x}_{1,i}$, $i = 1, 2, 3$ shown in Table 4b). This variable clearly reflects the improvement of confidence during the second half of the estimation period and suggests that this helped to reduce nominal volatility. In contrast, the $D^{x*}_{1,i}$ values shown in table 4a, which allow for the possible effect of f^* on US macroeconomic volatility are small and insignificant, suggesting that US macroeconomic volatility was not affected by the financial factor over this period. This is consistent with the US macro residuals shown in the lower left panel of Figure 1 which appear to be homoscedastic, with very large negative residuals for the output gap in October and November 2008. The effect of residual sovereign bond market risks is captured by q_t which following previous studies we interpret as a liquidity premium. This factor is strongly correlated with the associated spreads.

4.2 The macroeconomic models

These macro–yield systems are analyzed in terms of their impulse responses, which show the dynamic effects of innovations in the latent and macroeconomic variables on the macro VAR systems. These responses depend upon the ordering of the variables in the state space vector. We have followed the convention of ordering factors first, followed the output gap, inflation and interest rates, with global/US variables ordered before Brazilian ones. Since the factors are non-stationary, their shocks have permanent effects. The output gap is mean-reverting

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and has temporary effects, irrespective of the variable ordering. Inflation and interest rates are non-stationary, but with this ordering the impulse responses show the effect of conditional shocks or deviations from the non-stationary central tendency determined by z^*, f^* and f, which are also mean reverting. In other words, the long run effects of inflation and interest rates are picked up by the associated factors.

The impulse responses of the US macro KVAR is shown in Figure 4. These are broadly in line with results from previous US single country macro-finance models. The real factor z^* behaves like a demand shock. It is correlated with real output, inflation and interest rates in the short run, but by construction has a unit impact on the US interest rate in the steady state. The financial factor f^* acts like a supply shock, reducing output temporarily but increasing inflation and interest rates, with a unit effect on both in steady state. Output and inflation shocks have plausible effects on the rest of the system. As expected, increases in output and inflation trigger an increase in the policy rate.

The impulse responses of the Brazilian macro KVAR are shown in Figure 5. Because this is an $N = 1$ model, these responses are smoother than for the US. The steady state common factor and US spillover coefficients are reported in Table 4b. Most of the work is done by the common trends z^* and f^*. Their effects in Brazil are consistent with their role as global demand and supply shocks, exhibiting a pattern similar to that of the US model shown in Fig 4. The US business cycle (g^*) has significant spillover effects and the inflation effect is also of significance. Conditional upon these effects, the US policy rate has no direct effect on Brazil. However, an increase in the US policy rate tends to reduce Brazilian output and inflation indirectly, hence reducing the Brazilian policy rate. The lower panel of Figure 5 shows the effect of Brazil-specific shocks. A general deterioration in confidence reflected in f leads

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to persistently higher inflation and interest rates but apparently increases output temporarily, an effect that could be due to currency depreciation. As in the case of the US, output and inflation shocks have plausible effects on the rest of the system while a rise in the policy rate depresses output and inflation.

4.3 The sovereign spread model

The impulse responses for the US Treasury market are shown in Figure 6. These are broadly in line with what we would expect from previous macro-finance studies, although f^* has a negative effect on the medium and long run maturity yields, perhaps reflecting 'flight to quality' effects. Since z^* and f^* are both non-stationary in this data set, they have persistent effects on the curve, acting as 'level' factors. As usual, the policy interest rate acts as a slope factor and the output gap as a curvature factor.

The main objective of this paper is however to obtain measures of the two Brazilian financial factors and their effects and those of the macro variables on the sovereign spreads. The next equation ((20), with the parameters reported in Table 4c) shows how the short sovereign bond spread is influenced by these various effects[7]:

$$\begin{aligned} s_t = f_t + q_t + 0.68655\pi_t - 0.37821 r_t + 0.25223 r_t^* \\ (2.71) \qquad (3.51) \qquad (1.79) \end{aligned} \tag{26}$$

This equation suggests that macro effects are significant in the most recent data. It suggests that spreads are positively correlated with US interest and the Brazilian inflation rates but

[7] Given the effects of the Brazil output gap variable is negligibly small, we don't include it in the spread equation for the final model M1.

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negatively correlated with the Brazilian policy rate. These results seem plausible. An increase in inflation presumably increases default risk while an increase in the policy interest rate will help to contain this risk. A rise in the US interest rate tends to increase the burden of debt interest payments for sovereign borrowers and could reduce the market's appetite for risky assets.

The impulse responses for the longer maturity spreads depend upon the way in which the macroeconomy and confidence (reflected in the risk premia) are affected by shocks to the system. The top panel in Figure 7 shows how the sovereign spreads respond to global/US shocks. A positive global demand shock reflected in the real factor z^* reduces the 5 and 10 year spreads, improving investor confidence. An increase in f^* initially increases the short sovereign spread through the pronounced effect it has on the Brazilian policy rate, but reduces long term spreads through its effect on confidence. Increases in US output and inflation reduce spreads while US interest rates increase short and medium term spreads. However as we have noted, these conditional influences are mean-reverting and their effects are not persistent. The lower panel in Figure 7 shows how the sovereign spreads respond to the Brazil-specific shocks. They suggest that an increases in both f and q lead to persistently higher sovereign bond spreads. This is largely due to the direct effect of the deterioration in confidence but partly due to the indirect effect of higher output and inflation. Brazilian output, inflation and interest rates all reduce the medium and long term sovereign spreads but the effects are again short-lived.

Figure 8 shows the factor loadings or coefficients of the spread model (25) as a function of maturity expressed in months. The Brazilian financial factor f_t naturally dominates the behavior of the spreads during periods of crisis. The loading of the liquidity risk factor q_t is small compared to that of f_t. However, the loadings of the macro variables are comparable

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with those of US macro-finance models (compare for example Figure 8 of Dewachter, Lyrio, and Maes (2006))). Recall that the long run effects of the inflation and interest rates is picked up by the financial factors, which is why their loadings decay towards zero at the long end.

5 Conclusion

While the recent global financial crisis provided a vivid demonstration of the links between different economies and financial markets, the growth of the BRICs and other emerging economies is now underlining their importance for the global recovery. The specification developed in this paper shows that macro-finance models provide a useful framework for exploring these international economic linkages and their implications for financial markets. It finds evidence of global trends in the US and Brazilian economies and financial markets as well as spillovers effects from US output and inflation to the Brazilian economy. The US policy rate affects spreads in the Brazilian sovereign debt markets, as do Brazilian inflation and policy rates. The Brazilian financial common risk factor naturally dominates the behavior of the spreads during periods of crisis and we find that it also affects the level and volatility of macroeconomic variables. However, its influence diminishes considerably after the crisis of 2002, leaving a more stable system in which macroeconomic influences come into play. Indeed, the role played by macroeconomic variables in the Brazilian sovereign markets appears to be similar to the role that they play in macro-finance models of the US Treasury bond market. These results suggest that the macro-finance approach could throw light upon the behavior of other emerging economies and markets.

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A Appendix

A.1 The companion form of the macro models

The US-Brazilian model is given as

$$\mathbf{X}_t = \mathbf{K} + \mathbf{\Phi}\mathbf{X}_{t-1} + \mathbf{W}_t \tag{27}$$

$$\mathbf{W}_t \sim N(\mathbf{0}, \mathbf{\Omega}_t) \tag{28}$$

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where[8]

$$\mathbf{W}_t = \mathbf{C}\mathbf{D}_{t-1}\boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_t \sim N(\mathbf{0}, \mathbf{I}); \boldsymbol{\Omega}_t = \mathbf{C}\mathbf{D}_{t-1}^2\mathbf{C}';$$

$$\mathbf{D}_t^2 = \mathbf{D}_0^2 + \mathbf{D}_1^2\mathbf{J}_{f*}\mathbf{X}_t + \mathbf{D}_2^2\mathbf{J}_f\mathbf{X}_t + \mathbf{D}_3^2\mathbf{J}_q\mathbf{X}_t$$

$$\mathbf{K} = \begin{pmatrix} 0 \\ 0 \\ \boldsymbol{\kappa}^* \\ \mathbf{0} \\ 0 \\ \boldsymbol{\kappa} + \boldsymbol{\theta}_{x1}\boldsymbol{\kappa}^* \\ \kappa^q \end{pmatrix}, \ \mathbf{X}_t = \begin{pmatrix} z_t^* \\ f_t^* \\ \mathbf{x}_t^{*\prime} \\ \mathbf{x}_{t-1}^{*\prime} \\ f_t \\ \mathbf{x}_t' \\ q_t \end{pmatrix}, \mathbf{J}_{f*} = \begin{pmatrix} 0 \\ 1 \\ \mathbf{0}_{11,1} \end{pmatrix},$$

$$\mathbf{J}_f = \begin{pmatrix} \mathbf{0}_{8,1} \\ 1 \\ \mathbf{0}_{4,1} \end{pmatrix}, \mathbf{J}_q = \begin{pmatrix} \mathbf{0}_{12,1} \\ 1 \end{pmatrix},$$

$$\boldsymbol{\Phi} = \begin{pmatrix} 1 & 0 & \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & 0 \\ 0 & \xi^{f*} & \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & 0 \\ \boldsymbol{\phi}_z^* & \boldsymbol{\phi}_f^*\xi^{f*} & \boldsymbol{\Phi}_1^* & \boldsymbol{\Phi}_2^* & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 0 & 0 & \mathbf{0} & \mathbf{0} & \xi^f & \mathbf{0} & 0 \\ (\boldsymbol{\theta}_z + \boldsymbol{\theta}_{x1}\boldsymbol{\phi}_z^*) & (\boldsymbol{\theta}_f + \boldsymbol{\theta}_{x1}\boldsymbol{\phi}_f^*)\xi^{f*} & (\boldsymbol{\theta}_{x1}\boldsymbol{\Phi}_1^* + \boldsymbol{\theta}_{x2}) & \boldsymbol{\theta}_{x1}\boldsymbol{\Phi}_2^* & \phi\xi^f & \boldsymbol{\Phi}_1 & \mathbf{0} \\ 0 & 0 & \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & \xi^q \end{pmatrix}$$

[8] We use $\mathbf{0}_{i,j}$ to represent an i by j zero matrix.

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$$
\mathbf{C}=\begin{pmatrix}
1 & 0 & \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & 0 \\
0 & 1 & \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & 0 \\
\boldsymbol{\phi}_z^* & \boldsymbol{\phi}_f^* & \mathbf{G}^* & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\
\mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\
0 & 0 & \mathbf{0} & \mathbf{0} & 1 & \mathbf{0} & 0 \\
(\boldsymbol{\theta}_z+\boldsymbol{\theta}_{x1}\boldsymbol{\phi}_z^*) & (\boldsymbol{\theta}_f+\boldsymbol{\theta}_{x1}\boldsymbol{\phi}_f^*) & (\boldsymbol{\theta}_{x1}\mathbf{G}^*) & \mathbf{0} & \boldsymbol{\phi}_f\,\mathbf{G} & \mathbf{0} & \\
0 & 0 & \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & 1
\end{pmatrix},
$$

$$
\mathbf{D}_0 = \begin{pmatrix}
\delta^{z*} & 0 & \mathbf{0}_{1,3} & \mathbf{0}_{1,4} & \mathbf{0}_{1,3} & 0 \\
0 & 0 & \mathbf{0}_{1,3} & \mathbf{0}_{1,4} & \mathbf{0}_{1,3} & 0 \\
\mathbf{0}_{3,1} & \mathbf{0}_{3,1} & \mathbf{D}_0^{x*} & \mathbf{0}_{3,4} & \mathbf{0}_{3,3} & \mathbf{0}_{3,1} \\
\mathbf{0}_{4,1} & \mathbf{0}_{4,1} & \mathbf{0}_{4,3} & \mathbf{0}_{4,4} & \mathbf{0}_{4,3} & \mathbf{0}_{4,1} \\
\mathbf{0}_{3,1} & \mathbf{0}_{3,1} & \mathbf{0}_{3,3} & \mathbf{0}_{3,4} & \mathbf{D}_0^{x} & \mathbf{0}_{3,1} \\
0 & 0 & \mathbf{0}_{1,3} & \mathbf{0}_{1,4} & \mathbf{0}_{1,3} & 0
\end{pmatrix}, \mathbf{D}_1 = \begin{pmatrix}
0 & 0 & \mathbf{0}_{1,3} & \mathbf{0}_{1,8} \\
0 & \delta^{f*} & \mathbf{0}_{1,3} & \mathbf{0}_{1,8} \\
\mathbf{0}_{3,1} & \mathbf{0}_{3,1} & \mathbf{D}_1^{x*} & \mathbf{0}_{3,8} \\
\mathbf{0}_{8,1} & \mathbf{0}_{8,1} & \mathbf{0}_{8,3} & \mathbf{0}_{8,8}
\end{pmatrix},
$$

$$
\mathbf{D}_2 = \begin{pmatrix}
\mathbf{0}_{8,8} & \mathbf{0}_{8,1} & \mathbf{0}_{8,3} & \mathbf{0}_{8,1} \\
\mathbf{0}_{1,8} & \delta^{f} & \mathbf{0}_{1,3} & 0 \\
\mathbf{0}_{3,8} & \mathbf{0}_{3,1} & \mathbf{D}_1^{x} & \mathbf{0}_{3,1} \\
\mathbf{0}_{1,8} & 0 & \mathbf{0}_{1,3} & 0
\end{pmatrix}, \mathbf{D}_3 = \begin{pmatrix}
\mathbf{0}_{12,12} & \mathbf{0}_{12,1} \\
\mathbf{0}_{1,12} & \delta^{q}
\end{pmatrix}
$$

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A.2 Modeling the price of risk and bond prices

In a standard affine model, the expected values of the state variables $\mathbf{X}_{t+1}$ under $\mathcal{P}$ and $\mathcal{Q}$ are related by market price of risks $\mathbf{\Lambda}_t$:

$$E_t^Q\left(\mathbf{X}_{t+1}\right) = E_t\left(\mathbf{X}_{t+1}\right) - \mathbf{C}\mathbf{D}_{t+1}\mathbf{\Lambda}_{t+1} \tag{29}$$

These equations explain the factor risk premia in terms of the prices of risk. For example (29) takes $\mathbf{\Lambda}_{t+1}$, the vector showing the excess return to standard deviation ratios associated with $\boldsymbol{\epsilon}_{t+1}$; multiplies these by the standard deviations $\mathbf{D}_t$ to get their differential drift and then maps these into the factor risk premia (the differential drift in $\mathbf{X}_{t+1}$) using $\mathbf{C}$. Following Duffee (2002), these prices of risk are affine in the state vector. For the US dollar investors, they price market risks as :

$$\mathbf{\Lambda}_{t+1} = \mathbf{D}_{t+1}\mathbf{\Lambda}_1 + \mathbf{D}_{t+1}^{-}\mathbf{\Lambda}_2\mathbf{X}_t \tag{30}$$

$$where\ :\ D_{t,ii}^{-} = \begin{cases} 0, \text{if i=2,9,13} \\ 1/D_{t,ii},\ \text{otherwise} \end{cases}$$

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where $\mathbf{D}_t$ is defined in (13). The parameters $\mathbf{\Lambda}_1$ and $\mathbf{\Lambda}_2$ are defined as:

$$\mathbf{\Lambda}_1 = \begin{pmatrix} \lambda_1^{z*} \\ \lambda_1^{f*} \\ \boldsymbol{\lambda}_1^{x*} \\ \mathbf{0}_{3,1} \\ \lambda_1^{f} \\ \boldsymbol{\lambda}_1^{x} \\ \lambda_1^{q} \end{pmatrix}, \mathbf{\Lambda}_2 = \begin{pmatrix} \lambda_2^{z} & 0 & \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & 0 \\ 0 & 0 & \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & 0 \\ \mathbf{0} & \mathbf{0} & \boldsymbol{\lambda}_2^{x*} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 0 & 0 & \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \boldsymbol{\lambda}_2^{x} & \mathbf{0} \\ 0 & 0 & \mathbf{0} & \mathbf{0} & 0 & \mathbf{0} & 0 \end{pmatrix}, \tag{31}$$

where $\boldsymbol{\lambda}_1^x, \boldsymbol{\lambda}_1^{x*}$ are 3 by 1 column vectors; $\boldsymbol{\lambda}_2^{x*}$ and $\boldsymbol{\lambda}_2^x$ are 3 by 3 matrices; other lambda parameters are scalers. Substituting (30) into (29) gives:

$$\mathbf{K}^Q = \mathbf{K} - \mathbf{C}\mathbf{D}_0^2\mathbf{\Lambda}_1$$

$$\mathbf{\Phi}^Q = \mathbf{\Phi} - \mathbf{C}\mathbf{D}_1^2\mathbf{\Lambda}_1\mathbf{J}_{f*} - \mathbf{C}\mathbf{D}_2^2\mathbf{\Lambda}_1\mathbf{J}_f - \mathbf{C}\mathbf{D}_3^2\mathbf{\Lambda}_1\mathbf{J}_q - \mathbf{C}\mathbf{\Lambda}_2$$

A.3 Kalman Filter estimation

The model is estimated using Extended Kalman Filter techniques and maximum loglikelihood method by using the state-space system:

$$\mathbf{X}_t = \mathbf{K} + \mathbf{\Phi}\mathbf{X}_{t-1} + \mathbf{W}_t, \mathbf{W}_t \sim N(\mathbf{0}, \mathbf{\Omega}_t) \tag{32}$$

$$\mathbf{Y}_t = \mathbf{h}_t(\mathbf{X}_t) + \mathbf{E}_t, \mathbf{E}_t \sim N(\mathbf{0}, \mathbf{Q}) \tag{33}$$

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The transition equation (32) is the companion form of the US-Brazil macro model under the real world measure $\mathcal{P}$. The measurement equation (33) consists of the US yield model (23) and Brazilian yield model (24), where all the models are based on the parameters defined under $\mathcal{Q}$. The observation vector $\mathbf{Y}_t$ can be a non-linear function of the state vector $\mathbf{X}_t$.

The observation vector $\mathbf{Y}_t$ includes observed macro and financial data:

$$\mathbf{Y}_t' = \{\mathbf{x}_t^{*\prime}, \mathbf{x}_t', \mathbf{y}_t^{*\prime}, \mathbf{y}_t']$$

where $\mathbf{y}_t^*$ is the US yield vector, $\mathbf{x}_t^*$ is the US observed macro vector, $\mathbf{y}_t$ is the Brazilian yield, and $\mathbf{x}_t$ is the observed Brazilian macro vector.

Also we have:

$$\mathbf{h}_t\left(\mathbf{X}_t\right) = \begin{pmatrix} \mathbf{x}_t^* \\ \mathbf{x}_t \\ \mathbf{y}^*(\mathbf{X}_t) \\ \mathbf{y}(\mathbf{X}_t) \end{pmatrix}, \mathbf{Q} = \begin{pmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{\Sigma}^* & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{\Sigma} \end{pmatrix}$$

where $\mathbf{y}^*(\mathbf{X}_t)$, $\mathbf{y}(\mathbf{X}_t)$ are affine functions of $\mathbf{X}_t$ to calculate the US yield function and Brazilian yield respectively. Please see Liu (2007) for more details of the Extended Kalman Filter techniques and maximum loglikelihood method.

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B Tables

Table 1

Summary statistics

	US and Brazilian macro variables					
	g^*	π^*	r^*	g	π	r
Mean	0.052	2.483	2.909	-0.364	6.090	17.502
Std.	4.726	1.335	1.930	3.673	2.695	5.153
Skew	-0.493	-0.955	0.116	-1.008	1.472	0.674
Kurt.	0.595	1.602	-1.361	3.026	3.574	1.125
ADF	-3.43*	-2.123	-2.625	-3.576*	-2.513	-1.831
	US yields					
	y_1^*	y_2^*	y_3^*	y_5^*	y_7^*	y_{10}^*
Mean	3.233	3.461	3.693	4.085	4.369	4.675
Std.	1.840	1.650	1.477	1.206	1.01	0.836
Skew	0.101	0.082	0.074	0.125	0.181	0.189
Kurt.	-1.282	-1.107	-0.959	-0.576	-0.218	0.035
ADF	-0.268	-0.585	-0.841	-1.219	-1.501	-1.848
	Brazil sovereign yields					
		y_2	y_3	y_5	y_7	y_{10}
Mean		7.488	8.384	9.544	10.327	11.164
Std.		4.566	4.625	4.611	4.587	4.602
Skew		2.314	1.991	1.616	1.404	1.226
Kurt.		8.475	6.284	4.128	2.994	2.052
ADF		-1.947	-1.905	-1.813	-1.715	-1.575

Mean denotes sample arithmetic mean expressed as percentage p.a.and $Std.$ standard deviation $Skew.$ & $Kurt.$ are standard measures of skewness (the third moment) and excess kurtosis (the fourth moment). ADF is the Adjusted Dickey-Fuller statistic testing the null hypothesis of non-stationarity (where the appropriate lag length is chosen using AIC). The ADF 10% and 5% significance levels are 2.575 and 2.877 respectively.

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Table 2

BIC test results for the basic macro models

Lag	1	2	3
US macro VAR	-1.99424	-2.18641	-1.94551
Brazil macro VAR	3.44125	3.66615	3.78816

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Table 3

Log likelihood ratio tests

Model (M)	Specification	Parameters k(M)	k(3)-k(M)	k(2)-k(M)	k(1)-k(M)	Loglike-lihood L(M)	LR tests against M3 2x(L(3)-L(M))	against M2 2x(L(2)-L(M))	against M1 2x(L(1)-L(M))
M0	Closed	88	24	17	8	1230.4	102.0	91.4	84.6
						χ^2_{95}	36.42	26.30	16.9
M1	Final open economy	96	16	9		1272.7	17.4	6.8	
						χ^2_{95}	25.00	14.07	
M2	Restricted open economy	105	7			1276.1	10.6		
						χ^2_{95}	15.51		
M3	Unrestricted open economy	112				1281.4			

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Table 4a

US macromodel parameter estimates (in Model M1)

Parameter	Estimate	t-value	Parameter	Estimate	t-value	Parameter	Estimate	t-value
$\Phi^*_{1,11}$	0.50917	6.11	$\Phi^*_{2,11}$	0.31079	3.90	G^*_{21}	0.03198	2.19
$\Phi^*_{1,12}$	−0.00012	0.00	$\Phi^*_{2,12}$	0.59456	1.56	G^*_{31}	0.01180	1.57
$\Phi^*_{1,13}$	0.00453	0.01	$\Phi^*_{2,13}$	−0.01097	−0.01	G^*_{32}	0.00000	0.00
$\Phi^*_{1,21}$	0.05221	3.39	$\Phi^*_{2,21}$	−0.02572	−1.66	$D^{x*}_{0,1}$	2.52792	4.59
$\Phi^*_{1,22}$	1.36223	17.40	$\Phi^*_{2,22}$	−0.50816	−6.83	$D^{x*}_{0,2}$	0.42760	4.59
$\Phi^*_{1,23}$	0.08749	0.51	$\Phi^*_{2,23}$	−0.10138	−0.62	$D^{x*}_{0,3}$	0.21580	4.58
$\Phi^*_{1,31}$	0.00000	0.00	$\Phi^*_{2,31}$	0.00000	0.00	$D^{x*}_{1,1}$	0.00000	0.00
$\Phi^*_{1,32}$	0.00296	0.10	$\Phi^*_{2,32}$	0.03165	1.19	$D^{x*}_{1,2}$	0.00001	0.00
$\Phi^*_{1,33}$	1.01389	14.60	$\Phi^*_{2,33}$	−0.08175	−1.28	$D^{x*}_{1,3}$	0.00003	0.00
φ_1	0.00000	0.00	δ^*_z	0.86855	4.78			
φ_2	0.00026	0.00	δ^*_f	0.12836	5.64	$[\xi^{f*}]$	1	[−]

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Table 4b

Brazilian macromodel parameter estimates

Parameter	Estimate	t-value	Parameter	Estimate	t-value	Parameter	Estimate	t-value
$\Phi_{1,11}$	0.74412	12.05	χ_{31}	0.47065	1.92	$D^x_{0,1}$	1.82809	9.33
$\Phi_{1,12}$	0.00033	0.00	χ_{22}	0.89629	1.45	$D^x_{0,2}$	0.62157	6.04
$\Phi_{1,13}$	−0.03251	−1.37	χ_{32}	2.31561	1.30	$D^x_{0,3}$	0.19615	1.91
$\Phi_{1,21}$	0.00634	0.25	χ_{13}	0.92193	3.00	$D^x_{1,1}$	0.00000	0.00
$\Phi_{1,22}$	0.84924	15.71	χ_{33}	1.44124	2.51	$D^x_{1,2}$	1.05824	4.75
$\Phi_{1,23}$	0.00000	0.00	χ_{24}	0.27106	0.76	$D^x_{1,3}$	1.72195	5.30
$\Phi_{1,31}$	0.09950	3.08	χ_{34}	1.46872	1.33	G_{21}	0.15275	2.88
$\Phi_{1,32}$	0.16417	3.07	χ_{16}	−0.30847	−1.04	G_{31}	0.04650	1.18
$\Phi_{1,33}$	0.90042	35.95	χ_{36}	−1.26028	−2.44	G_{32}	0.17405	2.39
φ_1	0.07739	0.03				δ_f	0.14665	9.00
φ_2	0.01457	0.00	υ	3.30843	1.78	$[\xi^f]$	1	[−]

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Table 4c

Estimates of the price of risk parameters

Parameter	Estimate	t-statistic	Parameter	Estimate	t-statistic
λ_1^{z*}	-0.09786	-1.37	λ_2^{z*}	0.04188	5.80
λ_1^{f*}	0.16991	3.24	$\lambda_{2,11}^{*}$	-0.18297	-1.00
λ_1^{g*}	-0.00040	0.00	$\lambda_{2,12}^{*}$	1.17588	1.78
$\lambda_1^{\pi*}$	3.51322	0.65	$\lambda_{2,13}^{*}$	-0.20696	-0.60
λ_1^{r*}	-6.86869	-1.72	$\lambda_{2,21}^{*}$	-0.02285	-0.41
λ_1^{f}	0.20021	0.22	$\lambda_{2,22}^{*}$	0.00000	0.00
λ_1^{g}	0.07118	0.15	$\lambda_{2,23}^{*}$	0.02209	0.20
λ_1^{π}	1.32405	1.25	$\lambda_{2,31}^{*}$	-0.01316	-1.51
λ_1^{r}	1.06560	1.33	$\lambda_{2,32}^{*}$	0.00910	0.30
λ_1^{q}	0.04005	2.90	$\lambda_{2,32}^{*}$	0.04583	2.77
			$\lambda_{2,11}$	0.00020	0.00
			$\lambda_{2,12}$	-0.13193	-0.33
κ^q	0.00635	0.36	$\lambda_{2,13}$	0.00000	0.00
δ^q	0.77911	8.22	$\lambda_{2,21}$	-0.00785	-0.02
$[\xi^q]$	1	$[-]$	$\lambda_{2,22}$	-0.12980	-1.42
			$\lambda_{2,23}$	0.00000	0.00
γ^{r*}	0.25223	1.79	$\lambda_{2,31}$	-0.00002	0.00
γ^{π}	0.68655	2.71	$\lambda_{2,32}$	-0.00071	0.00
γ^{r}	-0.37821	-3.51	$\lambda_{2,33}$	0.00000	0.00

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Table 5

The central tendency equations for the Brazilian economy in model M1

effect: on:	z_t^*	f_t^*	g_t^*	π_t^*	r_t^*	g_{t-1}^*	π_{t-1}^*	r_{t-1}^*	f_t
g_t	[0]	[0]	0.9219 (3.00)	[0]	[0]	−0.3085 (−1.04)	[0]	[0]	[0]
π_t	[0]	0.8963 (1.45)	[0]	0.2711 (0.76)	[0]	[0]	[0]	[0]	3.3084 (1.78)
r_t	0.4707 (1.92)	2.3156 (1.30)	1.4413 (2.51)	1.4687 (1.33)	[0]	−1.2603 (−2.44)	[0]	[0]	3.3084 (1.78)

This table shows the empirical values of the slope coefficients shown in Equation (7). The intercepts φ_1 and φ_2 are very close to zero and are not reported. The steady state common factor and spillover effects are represented by the central tendency Ξ defined in Equation (9). Insignificant effects are eliminated in M1. In this paper restrictions are indicated using square brackets as [0]. The numbers in round brackets are t-statistics.

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C Figures

Figure 1
Macro Variables

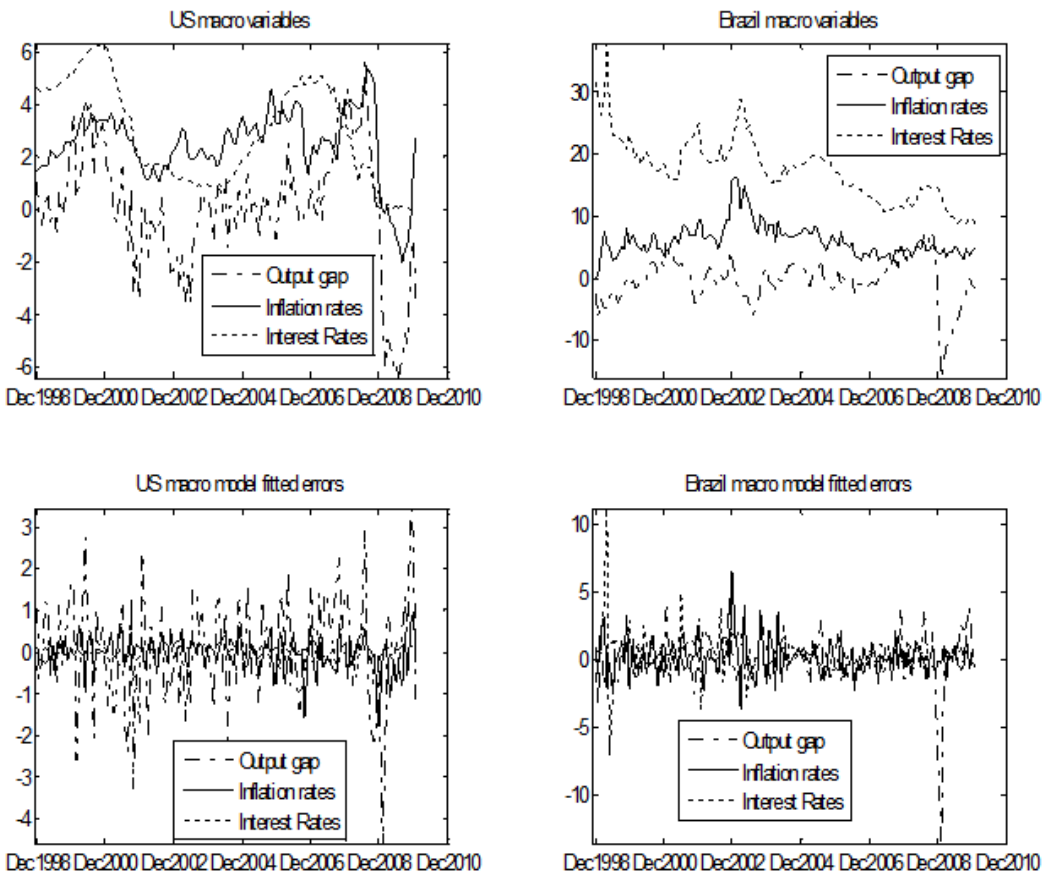

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Figure 2

US Treasury yields and Brazilain sovereign yields

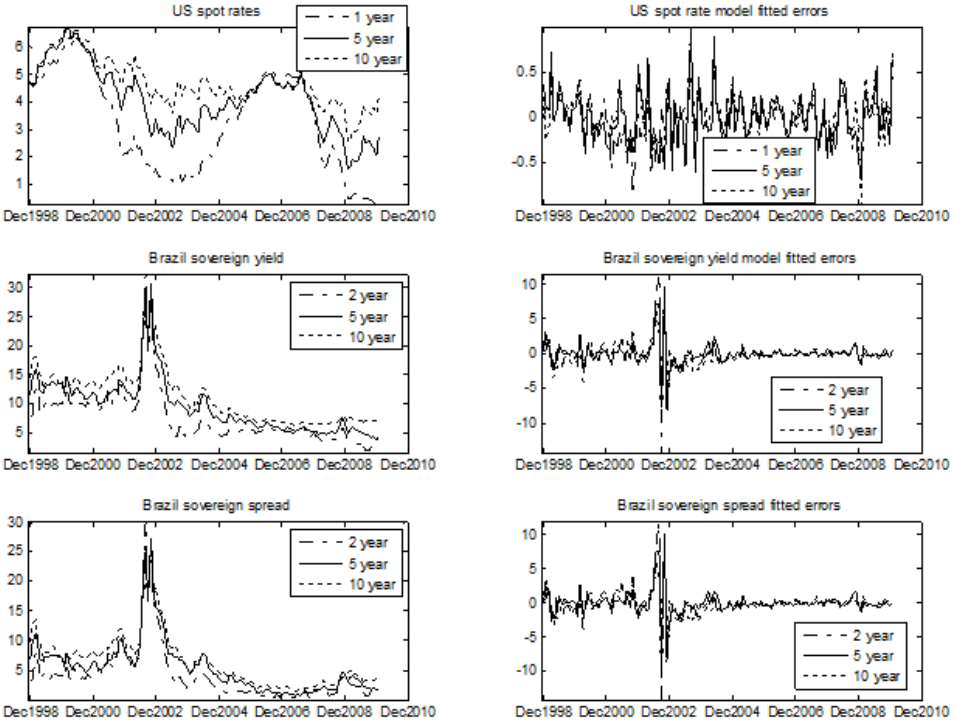

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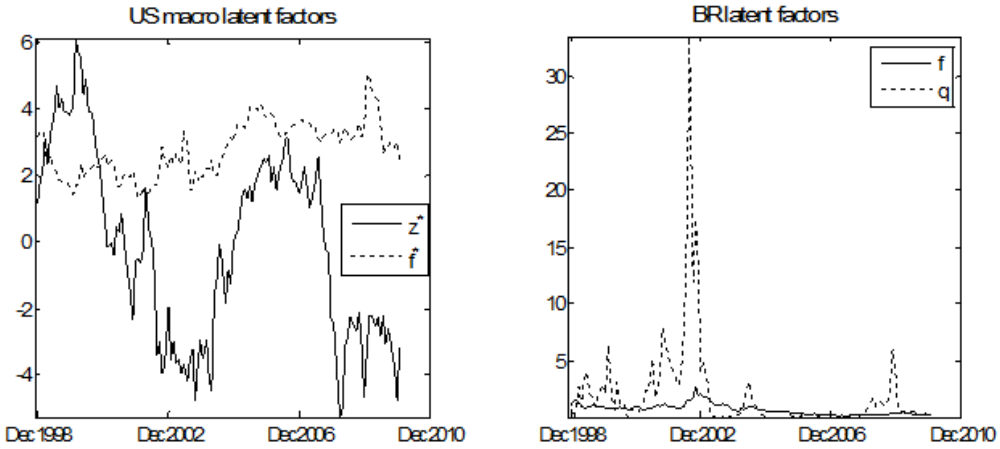
Figure 3
Latent Factors



These factors are estimated by the Extended Kalman Filter as described in Appendix A.3. In this data set they are all non-stationary. They are multiplied by a factor of 400 to make them consistent with percentage changes. z*, f and f* determine the central tendencies of the macro systems via (4) and (7). Q represents the residual of market-specific influences on Brazilian sovereign bond.

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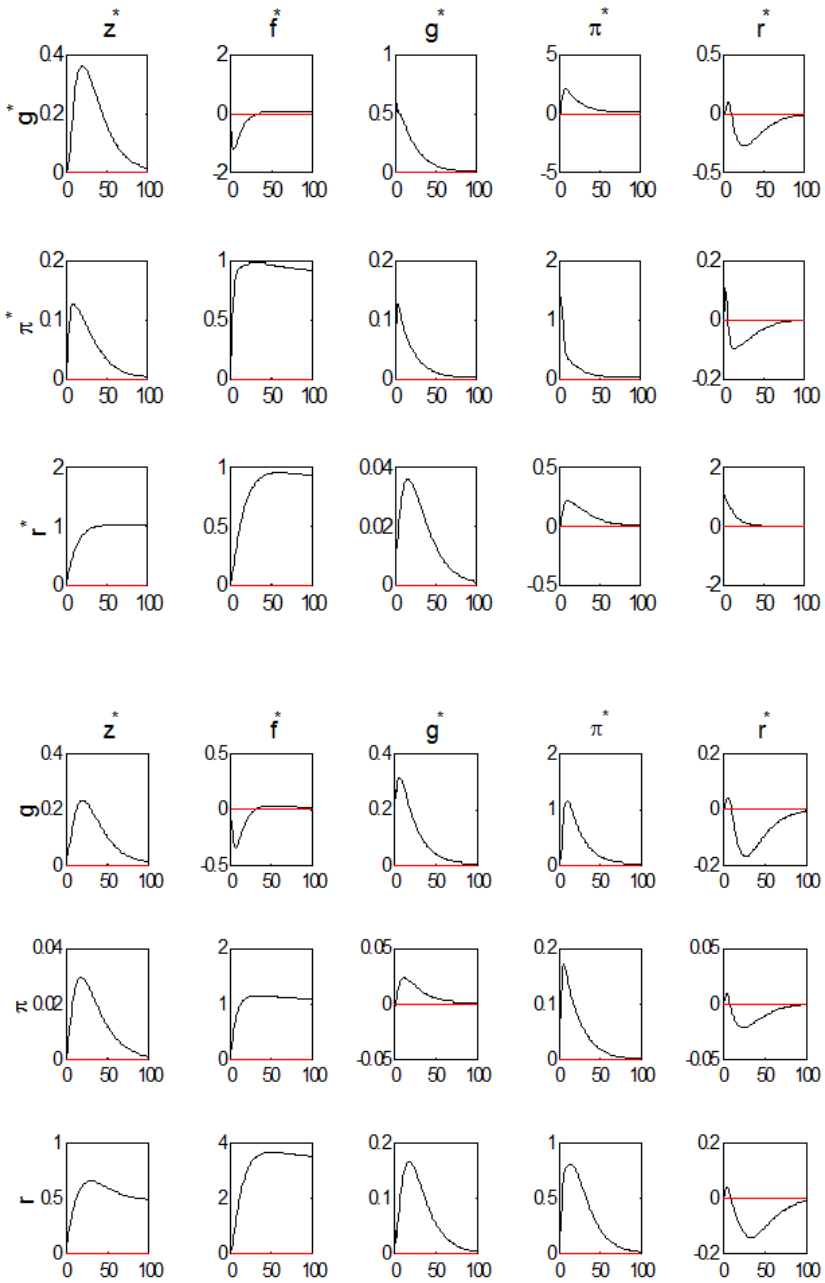
Figure 4

Impulse responses for the US macro KVAR



Each column shows the effect of a shock to one the orthogonal innovations (ϵ_t) shown in (12). These shocks increase the each of the driving variables in turn by one percentage point compared to its historical value for just one month. Since z^* and f^* are I(1), their disturbances have a permanent effect on inflation and interest rates. Other shocks are conditionally I(0) and thus have transient effcts. Elapsed time is measured in months.

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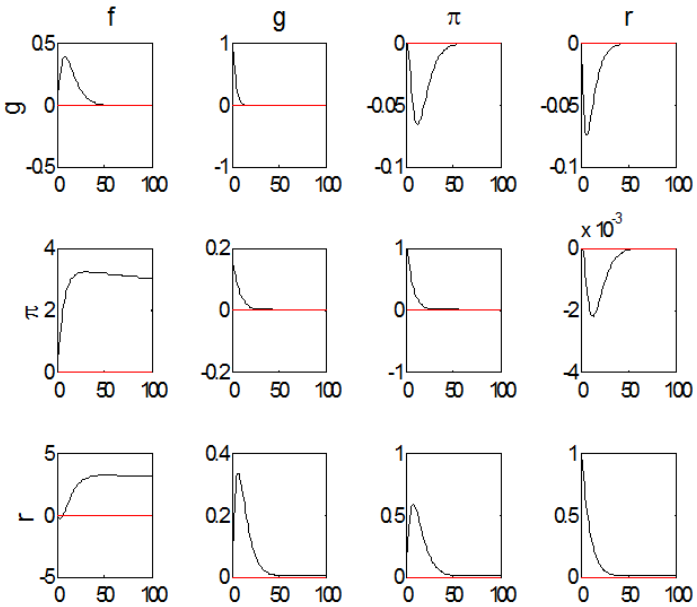
Figure 5

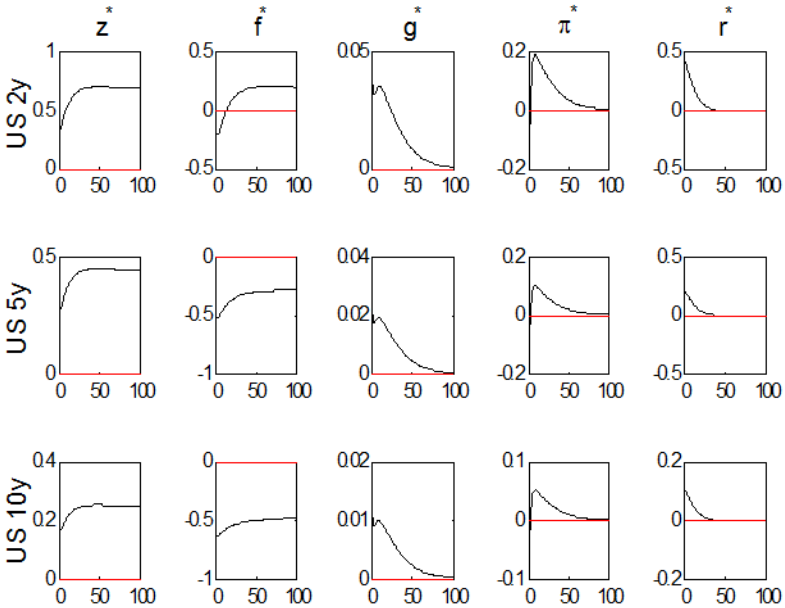
Impulse responses for theBrazilian macro KVAR



See footnotes to figure 4.

Figure 6

Impulse responses for the US yield model



See footnotes to figure 4.

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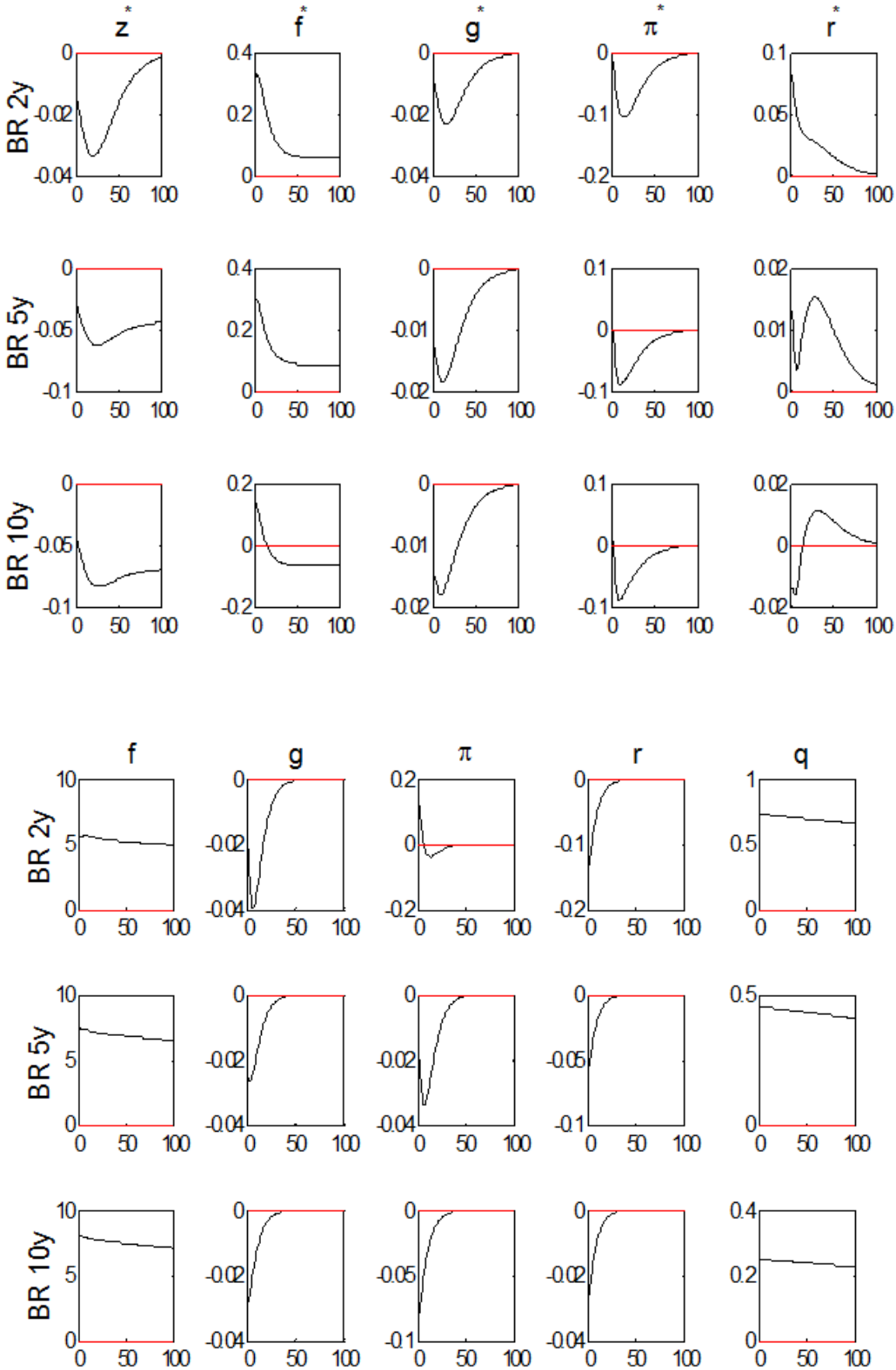
Figure 7

Impulse responses for the Brazilian sovereign spread model



See footnotes to figure 4. The spreads are determined by equation (25).

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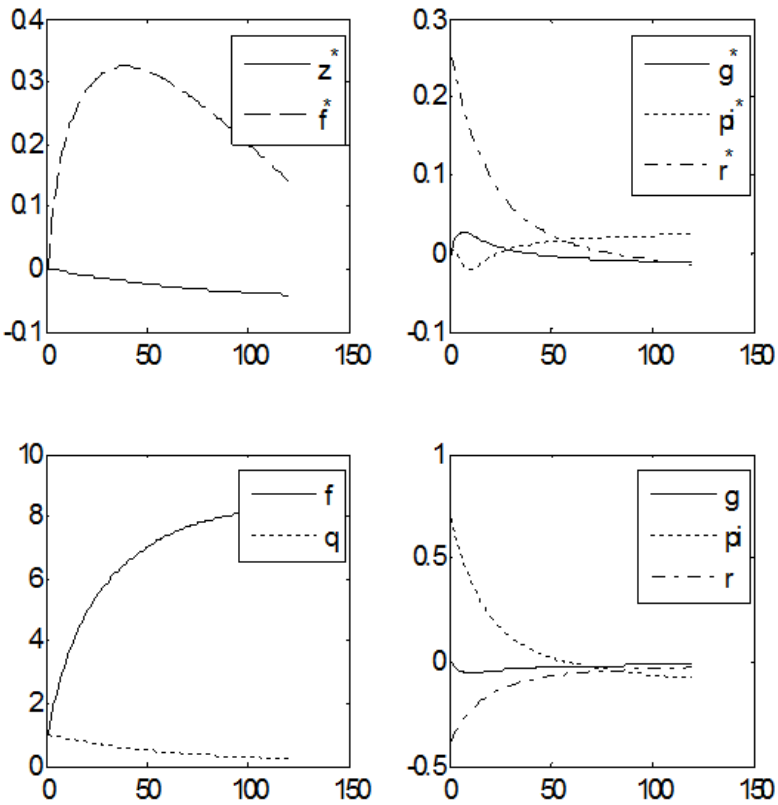
Figure 8

Factor loadings of the Brazil sovereign bond spread equation



The spreads are determined by equation (25): $s_{\tau,t} = y_{\tau,t} - y^*_{\tau,t} = \alpha_\tau - \alpha^*_\tau + (\boldsymbol{\beta}_\tau - \boldsymbol{\beta}^*_\tau) X_t$. The factor loadings are the slope coefficients in this equation.

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